Production of ε-Caprolactam Using Phenol Derived from Biomass

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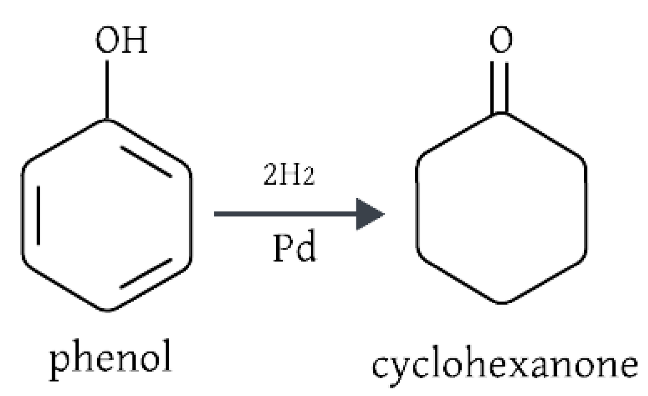
Mission

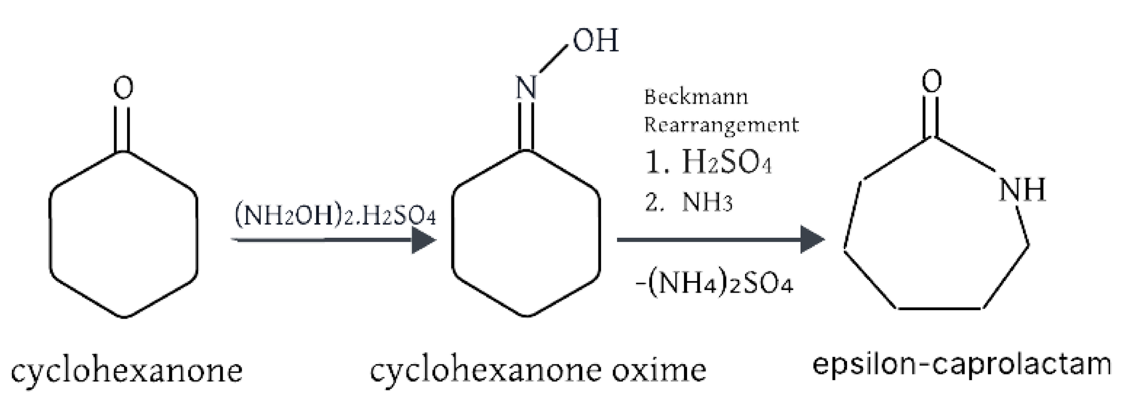
- Design a chemical plant located in New Mexico to produce sustainable ε-caprolactam from biomass-derived phenol.
- Purify ε-caprolactam to 98.5% [1] and produce 500,000 tons per year.

Research

- Epsilon-caprolactam is used make nylon 6, which is used to make textiles such as clothing and carpet.
- Phenol comes from lignin found in biomass.
- Biomass-derived phenol feedstock provides a renewable alternative to petroleum-based production in nylon manufacturing, reducing fossil fuel dependence and emissions.
- Production of ε-caprolactam from cyclohexanone is well known.

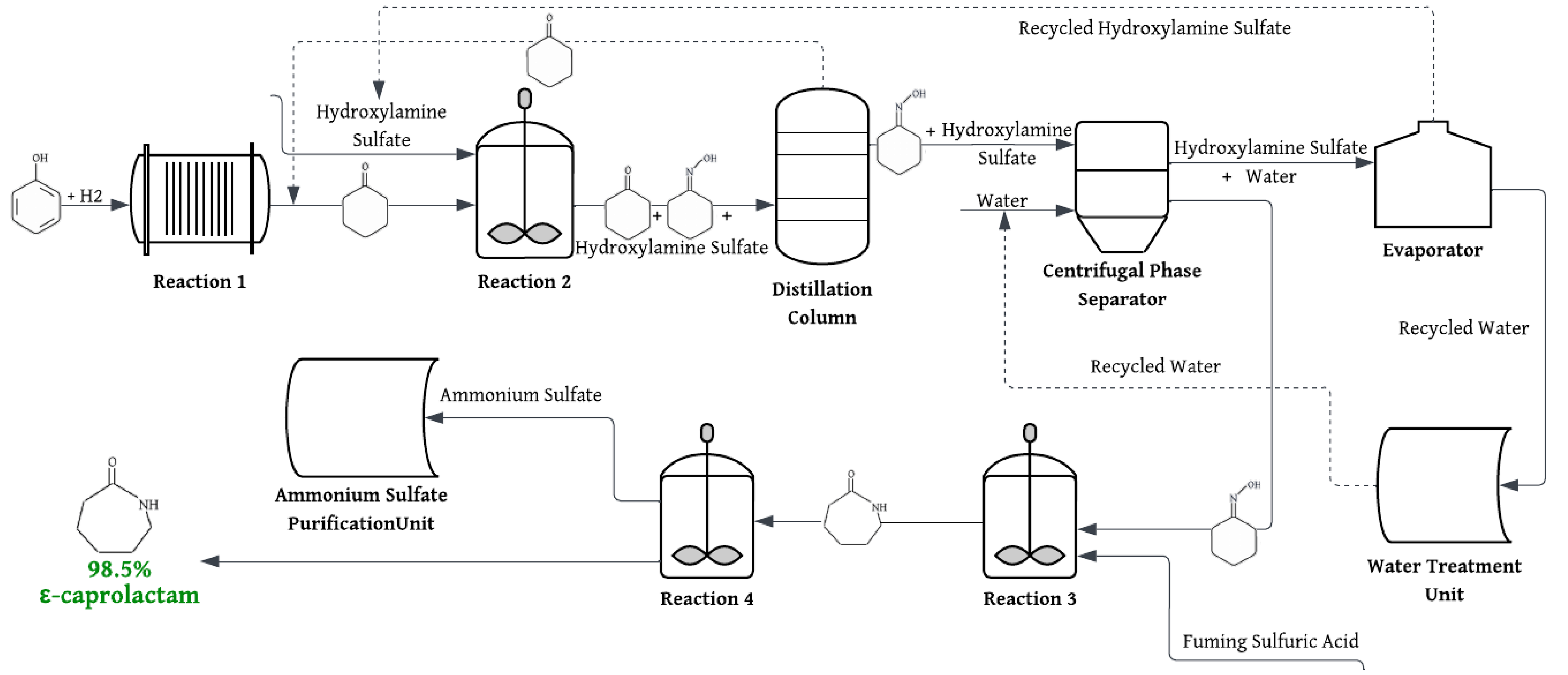


- Palladium ($15,000/kg) [3] and nickel catalysts ($85/kg) [4] can be used in the first reaction. Palladium catalyst has higher selectivity [5].

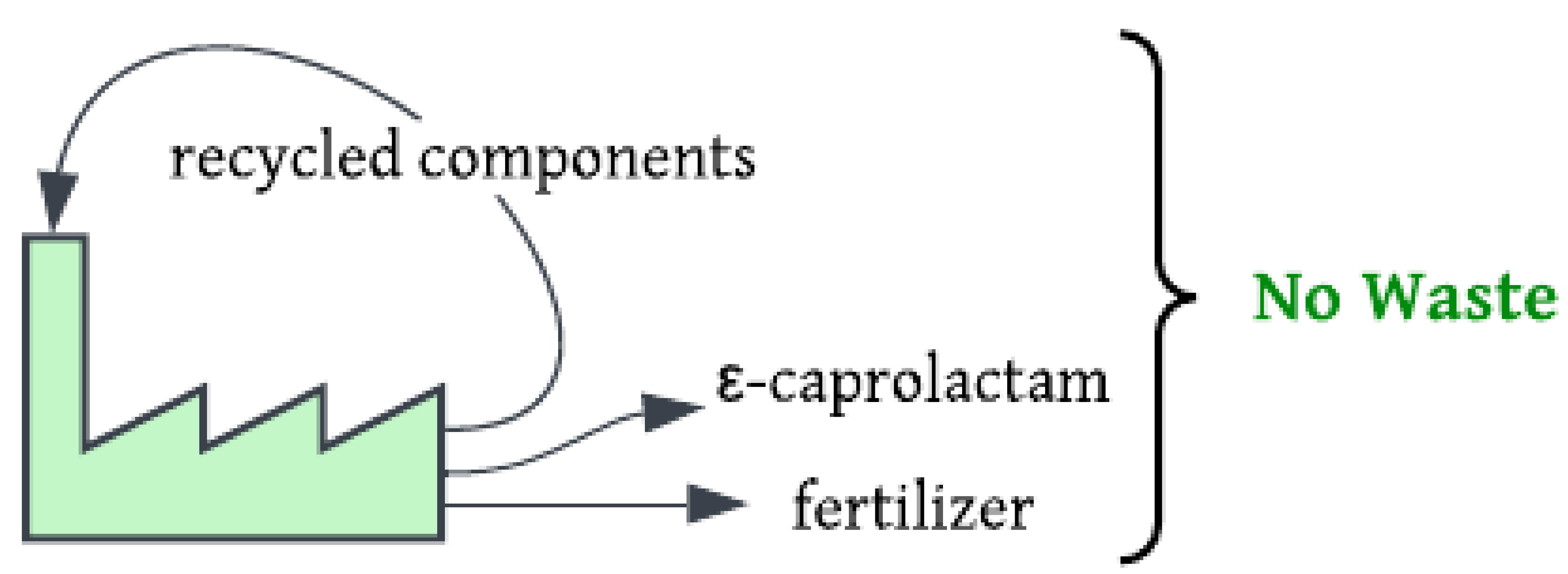


- A heat exchanger network reduced utility usage by 60%.
- Stainless steel is the material of construction to avoid corrosion and withstand high temperatures.

Final Design



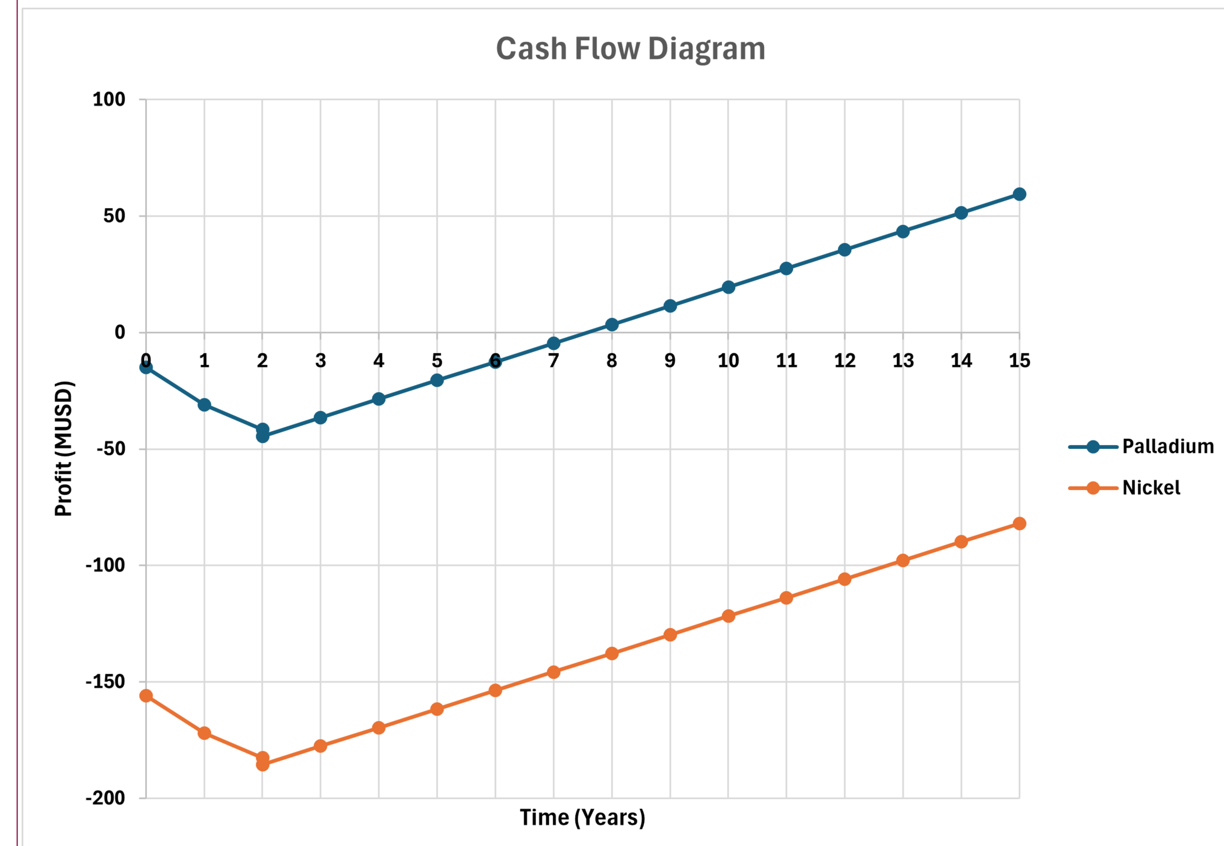
- Facility will be in Albuquerque, NM 87120, ensuring proximity to I-40 for raw material delivery, product distribution, and employee convenience.
- A plug flow reactor design for the first reaction allows for high conversion, continuous flow, and catalytic packing [6].
- Continuous stirred tank reactors for other reactions allow sufficient mixing, ensure uniform concentration and temperature, and minimize potential side reactions.
- A distillation column separates unreacted cyclohexanone from the first reaction for recycling.



- Ammonium sulfate is a co-product, which is purified and sold as a fertilizer.
- A centrifuge separates components from second reaction based on solubility – hydroxylamine sulfate is soluble in water while cyclohexanone oxime is not. Water and hydroxylamine sulfate are then also recycled.

Team Recommendations

- Management should proceed with process implementation - the proposed process is technically and economically feasible, producing 544,000 tons per year.
- Choose palladium over nickel catalyst. It costs more per kilogram, but its high selectivity to cyclohexanone [2] which increases efficiency, reduces amount of raw material and downstream processing needed.



Net Present Values of the process after 15 years for palladium and nickel catalysts are $19.3 M and -$77.5 M, respectively.

- Conduct pilot study to determine optimal reactor pressures.
- Quantify emissions for this process and compare to petroleum emissions as data to help investment decisions.

References

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